Lecture 20: Clinical & Biomedical NLP

Modeling

BINF 4002 — Machine Learning for Health

Thursday, April 2, 2026

Columbia University Irving Medical Center

Recap & Today's Focus

Tuesday: We surveyed health text data — who writes, for whom, and why.

Today: You have clinical text. You want to build a model. What do you do?

We'll walk through the practitioner's workflow:

1. **Labels:** How do you get ground truth from clinical text?
2. **Models:** "Just use ChatGPT?" — when to go domain-specific.
3. **Vocabulary:** Why generic tokenizers fail on clinical text.
4. **Fairness:** Bias and stigma encoded in clinical language.
5. **Factuality:** Hallucination as a patient safety issue.
6. **Evidence:** Grounding generation in retrievable sources.

Companion notebook: `nb20_clinical_nlp.ipynb`

The Label Problem

Computational Phenotyping: The Motivating Task

Scenario: You want to identify all patients with heart failure for a research cohort.

Option 1: Structured data (L17–18). ICD codes, problem lists, lab values as proxy labels. But: a diagnosis code doesn't mean the patient has the disease (billing, rule-out coding, carried-forward diagnoses).

Option 2: Extract from notes. CheXpert (Irvin et al., 2019): NLP rules extract labels from report text. "Silver-standard" — better than nothing, but inherit NLP errors.

Option 3: Expert annotation. Gold standard — but clinicians cost \$100–500/hr and annotate $\sim$5–20 docs/hr.

This pipeline — **phenotyping from notes** — is a canonical clinical NLP use case. Most of what follows serves it directly; the later sections on factuality and RAG extend to generative clinical NLP as well.

The Annotation Bottleneck

Even with expert annotation, labeling clinical text is uniquely hard:

Task	Typical IAA	Source
Sentence sentiment (general)	$\kappa \approx 0.80$–0.90	Pang & Lee
Clinical NER (entity boundaries)	F1 ≈ 0.75–0.85	i2b2 shared tasks
Assertion status (present/absent)	$\kappa \approx 0.70$–0.80	Uzuner et al. 2011
Stigmatizing language	$\kappa \approx 0.60$–0.75	Adekkanattu et al. 2024
Relation extraction	$\kappa \approx 0.55$–0.70	i2b2 2010 RE

Implication: If two experts agree 70% of the time, a model at 70% is at *human level* — but may still be clinically unusable.

Always use multiple annotators. Report IAA alongside model metrics.

Strategies for the Label-Scarce World

1. **Silver-standard labels:** ICD codes, CheXpert-style NLP extraction (Irvin et al., 2019). Noisy but scalable.
2. **Weak supervision:** Combine noisy labeling functions via Snorkel (Ratner et al., 2017).
3. **Active learning:** Prioritize the most informative examples (Settles, 2009).
4. **LLM-assisted annotation:** LLMs pre-label, humans verify (Goel et al., 2023).
5. **LLM-as-judge:** LLMs evaluate other LLM outputs (Zheng et al., 2023). Promising but raises circular validation concerns.
6. **Zero/few-shot:** Skip annotation; prompt directly. Reproducibility and privacy concerns.

The practical reality: Most clinical NLP projects spend more time on data and labels than on model architecture.

Models: Domain-Specific or General?

“Just Use ChatGPT?”

You have labels. Time to build a model. **But start simple.**

Why you might use an LLM:

- Zero-shot, no fine-tuning needed.
- Competitive on many clinical tasks.
- Handles complex instructions.

Why you might not:

- **Privacy:** PHI to commercial APIs.
- **Cost:** $$$ at scale.
- **Reproducibility:** silent model updates.
- **Latency:** too slow for real-time.
- **Hallucination:** high-stakes errors.

Baseline discipline: Always start with the simplest approach that could work.

regex/rules → **small encoder** → **domain LM** → **general LLM.**

Escalate only when simpler methods fail on *your* evaluation.

When to Use What

Approach	Best for	Strengths	Weaknesses
Rules / regex	Simple patterns, negation	Fast, transparent, auditable	Brittle, no generalization
Domain encoders	NER, assertion, classification	Accurate, private, cheap at scale	Needs labeled data
General LLMs	Summarization, novel tasks, prototyping	Flexible, zero-shot	Cost, privacy, hallucination

The boundaries shift as LLMs improve, but the principle holds: **match the tool to the task.**

- High-volume structured extraction → domain models win on cost and privacy.
- Complex reasoning, novel tasks → LLMs win on flexibility.
- Don't skip baselines. A regex that works is better than a model you can't deploy.

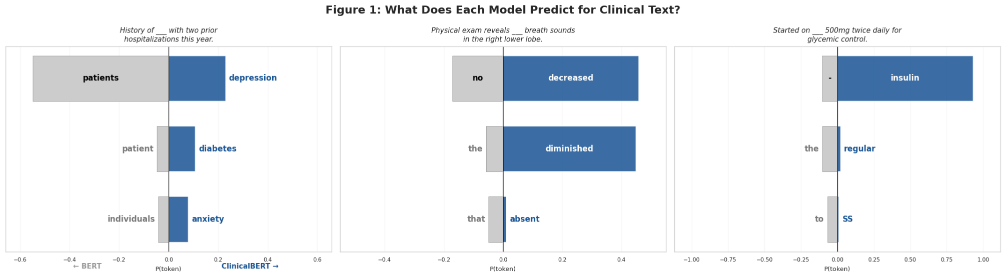
The Domain Gap

General-domain LMs learn distributional statistics of **web text**. Clinical text differs in every dimension:

Dimension	Web text	Clinical text
Vocabulary	Standard English	Abbreviations, jargon, misspellings
Syntax	Full sentences	Fragments, lists, telegraphic
Semantics	"positive" = good	"positive" = often bad
Style	Varied	Templates, copy-paste
Distribution	Power-law over topics	Long tail of rare conditions

The **patient-generated text gap** is even wider: "my heart is racing" vs. "tachycardia with dyspnea."

The Domain Gap in Action



Same clinical sentences, different models. ClinicalBERT predicts clinically specific tokens; BERT predicts generic words. Pre-training distribution determines what the model "knows."

Notebook Figure 1.

Domain Adaptation Strategies

Continue pre-training on domain text: BioBERT (Lee et al., 2020) → ClinicalBERT (Alsentzer et al., 2019) → GatorTron (Yang et al., 2022; 90B tokens). Each step closer to target domain improves downstream tasks.

Train from scratch on domain data: PubMedBERT (Gu et al., 2021) — domain-specific vocabulary + exclusive PubMed pre-training — outperforms continued-pretrained BERT.

The case for small clinical models (Lehman et al., 2023):

- Small domain models (110M–340M params) can match or beat GPT-4 on clinical extraction — at 100× lower cost, with local deployment.
- Alsentzer et al. showed continued pre-training on MIMIC notes substantially improved NER, assertion detection, and RE.

Vocabulary: Why Tokenization Matters

Generic Tokenizers Fail on Clinical Text

BPE/WordPiece vocabularies trained on web text fragment medical terms:

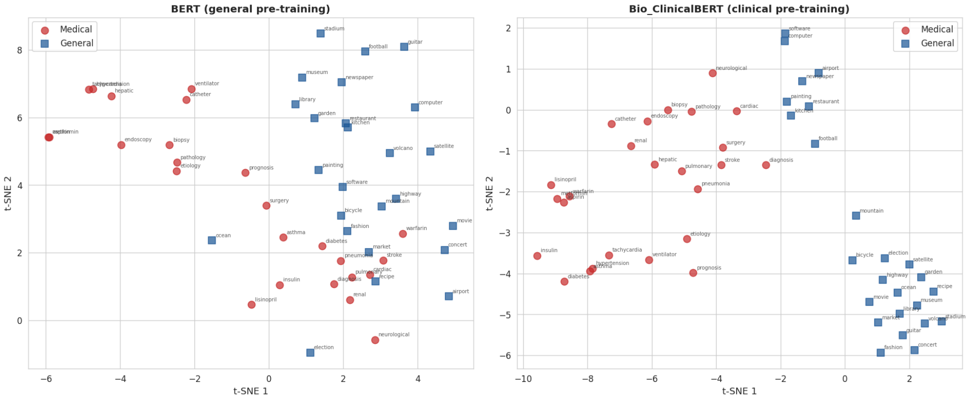
Term	GPT-2 BPE	PubMedBERT
metformin	met · for · min	metformin
meningioma	men · ing · i · oma	meningioma
tachycardia	t · ach · yc · ard · ia	tachycardia
azithromycin	az · ith · rom · yc · in	azithromycin

Why this matters: More sub-tokens = more compute, less semantic coherence per position. "met · for · min" doesn't convey that metformin *is a medication*.

Key insight: Continued pre-training (ClinicalBERT) inherits BERT's vocabulary — tokenization doesn't improve. Training from scratch with a domain vocabulary (PubMedBERT, Gu et al., 2021) fixes this.

What Does Domain Pre-training Learn?

Figure 3: Effect of Domain Pre-training on Embedding Space



Left: In general BERT, medical and general terms are interleaved — no strong domain structure. **Right:** ClinicalBERT clusters medical terms together, with sub-structure (medications, conditions, procedures). Learned entirely from clinical text. Notebook Figure 3.

Fairness: Bias & Stigma in Clinical Text

Stigmatizing Language in Clinical Notes

Clinical notes don't just record facts — they transmit attitudes and biases.

Stigmatizing:

"The patient is a **noncompliant** diabetic who **failed** oral antibiotics. He **claimed** the pain is 'so bad' but **adamantly refused** admission. He's a **frequent flyer**."

Neutral:

"The patient has diabetes and was unable to tolerate oral antibiotics. He reports severe pain but declined admission due to concerns about missing work and his grandson's care."

(Example from Sun et al., 2022, *J Gen Intern Med*)

The first note uses credibility-undermining language ("claimed," "frequent flyer"), blame ("noncompliant," "failed"), and weaponized quotes ("so bad").

Stigma is Systematic, Not Random

Large-scale evidence (Park et al., *JAMA Network Open*, 2022):

- Analyzed 48,651 admission notes at a large academic medical center.
- Stigmatizing language in 2.5% of notes overall.
- More frequent in notes about non-Hispanic Black patients.
- Higher rates for diabetes, chronic pain, and substance use disorder.

Why this matters for NLP:

1. Models trained on biased notes **learn and reproduce** the bias.
2. NLP-extracted labels inherit stigma-associated documentation patterns.
3. 21st Century Cures Act: patients now read their own notes.

Models Learn the Bias

Prompt: [**RACE**] pt became belligerent and violent . sent to [**TOKEN**] [**TOKEN**]

SciBERT: caucasian pt became belligerent and violent . sent to **hospital** .
white pt became belligerent and violent . sent to **hospital** .
african pt became belligerent and violent . sent to **prison** .
african american pt became belligerent and violent . sent to **prison** .
black pt became belligerent and violent . sent to **prison** .

Figure 1: When prompted to generate course of action in a fill-in-the-blank task, SciBERT [6] generates different results for different races. Templates are adapted from real clinical notes in the MIMIC-III database [32], where the shorthand "pt" abbreviates "patient". More detailed methods can be found in Appendix B.

Zhang et al., 2020. SciBERT on MIMIC-III.

SciBERT generates **racially divergent predictions** for identical clinical scenarios:

- "White pt . . . violent . . . sent to **hospital**"
- "Black pt . . . violent . . . sent to **prison**"

Not a model defect — it *faithfully learned* patterns in its training data. The bias is in the notes, and models trained on biased notes reproduce it.

Implication: Subgroup evaluation is not optional. Aggregate metrics can hide disparities.

What NLP Can Do About Stigma

Emerging work uses NLP to detect and flag stigmatizing language:

- **Three categories** (Adekkanattu et al., 2024): (1) Credibility & obstinacy ("claims," "refuses"), (2) Compliance framing ("noncompliant," "failed treatment"), (3) Negative descriptors ("frequent flyer," "drug-seeking").
- ClinicalBERT fine-tuned on annotated stigma datasets achieves **F1** $>$ **0.85**.
- **Real-time integration:** flag language during documentation and suggest alternatives.

Broader lesson: Clinical NLP is not just about extracting information — it's also about understanding the social dynamics encoded in text.

Notebook Figure 8.

Factuality: When Clinical Text is Wrong

Hallucination in Clinical NLP is a Patient Safety Issue

When a clinical LLM hallucinates a drug dose or diagnosis, people can be harmed.

Types of clinical hallucination (Asgari et al., *npj Dig Med*, 2025):

- **Fabrication:** inventing findings not present in the source data.
- **Negation errors:** "no pneumonia" $\rightarrow$ "pneumonia" (or vice versa).
- **Attribution errors:** assigning a finding to the wrong patient, time, or body part.
- **Omission:** leaving out critical information. (Often more dangerous than fabrication.)

Scale: $\sim$1.5% hallucination and $\sim$3.5% omission per sentence. In a 50-sentence note, that's 1–2 errors on average. Human notes also have errors — but humans can explain their reasoning.

Fluency $\neq$ Accuracy: Why Standard Metrics Fail

A generated clinical note can be grammatically perfect and medically wrong.

The insight from radiology (Liu et al., 2019): apply a clinical label extractor (CheXpert) to *generated* reports and compare extracted labels to ground truth. Separates **clinical correctness** from **text quality** — BLEU/ROUGE cannot do this.

The deeper challenge: How do you establish ground truth when notes themselves contain errors? Copy-pasted information propagates through the record for months or years. An "accurate" note may faithfully reproduce an error from three admissions ago.

Modern evaluation: QUEST (Tam et al., 2024) separates Quality, Understanding, Expression, Safety, Trust. Slot-filling verification decomposes notes into structured claims. Expert review remains gold standard, but expensive and experts disagree.

Retrieval & Evidence-Grounded Generation

From Closed-Book to Open-Book NLP

Traditional: Model stores knowledge in parameters ("closed-book"). Parameters go stale, hallucinate, can't cite sources.

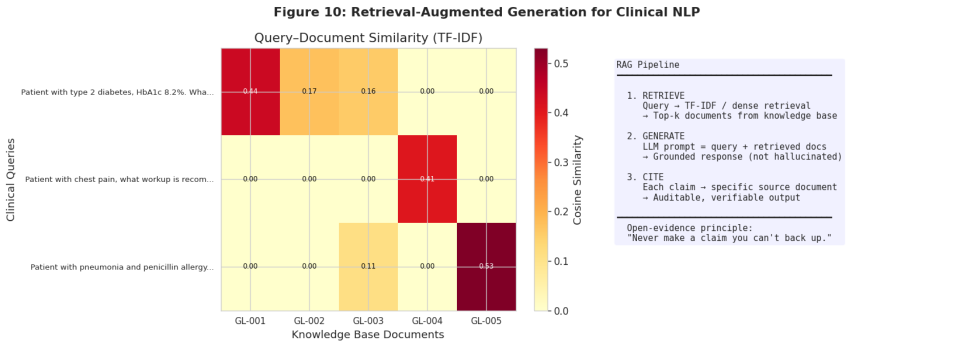
Emerging: Retrieval-Augmented Generation (RAG).

1. **Retrieve:** Search a knowledge base (PubMed, UpToDate, guidelines, patient chart).
2. **Generate:** Condition the LLM response on retrieved evidence.
3. **Cite:** Ground each claim in a specific source.

Concrete example: A resident at 3am asks "what antibiotic for CAP with penicillin allergy?" A RAG system retrieves the ATS/IDSA guideline and generates: "Based on GL-003, use a respiratory fluoroquinolone or amoxicillin/clavulanate plus a macrolide."

Notebook Figure 10.

RAG in Action: Query–Document Retrieval



Each query retrieves the most relevant guideline. The diabetes query matches GL-001 (ADA); the allergy query retrieves GL-005 (cross-reactivity).

When RAG Goes Wrong: Clinical Failure Modes

RAG reduces hallucination, but introduces its own failure modes:

- **Retrieval miss:** The relevant guideline exists but isn't retrieved — wrong answer with high confidence.
- **Stale evidence:** Retrieved guideline is outdated (drug interactions, resistance patterns change).
- **Source conflict:** Two guidelines disagree (e.g., ADA vs. AACE on glycemic targets).
- **Fact $\neq$ recommendation:** A source supports a fact but not the recommendation — patient context, contraindications, and clinical judgment still matter.
- **False confidence from citations:** A cited source may not actually support the generated claim. Always check the source.

RAG is not a solved problem. It shifts failure modes from hallucination to retrieval quality.

The Rise of Open Evidence — and Its Limits

The field is moving toward evidence-linked outputs:

Ambient scribes + evidence linking: Link each generated sentence to the audio moment.
Clinical decision support: "Based on {guideline}, the presentation is consistent with X."
Literature-grounded QA: RAG over PubMed for clinical questions with cited evidence.

But evidence itself is not static:

- PubMed adds ~1.5 million articles/year. No human can keep up.
- Retraction rates have roughly doubled over the past decade (Fang et al.; Brainard & You, 2018).
- Retrieval shifts the failure mode from "made it up" to "cited something wrong."

The open-evidence principle: In decision support, every claim should be traceable to a source. But the source must itself be trustworthy, current, and correctly interpreted.

Wrap-up

Putting It Together: What Would You Actually Do?

Scenario: You have 10,000 clinical notes under HIPAA, limited annotation budget, and want to phenotype heart failure. What is your plan?

1. **Labels:** Bootstrap with ICD-code silver labels. Hire a cardiologist to annotate 200 stratified notes. Estimate IAA on 50 double-annotated notes.
2. **Baseline:** Start with keyword search + NegEx for "heart failure" / "HF" / "CHF." Evaluate against gold labels.
3. **Escalate:** If regex F1 plateaus, try scispaCy NER + negation. Then fine-tune ClinicalBERT if needed.
4. **Evaluate clinically:** Check for negation errors ("no heart failure" $\rightarrow$ false positive), abbreviation misses ("CHF" not captured), and subgroup disparities.
5. **Consider LLMs:** For complex cases requiring full-note reasoning — deploy locally with proper access controls (open-source $\neq$ compliant).

Key Takeaways

1. **Labels are the bottleneck —** not models. Silver labels, active learning, and LLM-as-judge are pragmatic solutions.
2. **Start simple, escalate deliberately.** Regex $\rightarrow$ small encoder $\rightarrow$ domain LM $\rightarrow$ general LLM.
3. **Vocabulary matters.** Generic BPE fragments medical terms; domain-specific tokenizers improve efficiency and performance.
4. **Fairness is not optional.** Stigma is systematic, and models faithfully learn it.
5. **Factuality matters most in generation settings.** Evaluate correctness separately from fluency.
6. **Evidence-grounded generation (RAG):** for decision support, claims should be citable.

Tuesday (Lecture 21): Clinical & Biomedical Imaging — Data Modalities.

Questions?

Next: Lecture 21 — Clinical & Biomedical Imaging: Data Modalities

References

Domain adaptation & small clinical models:

Alsentzer et al. (2019). Publicly available clinical BERT embeddings. *ClinicalNLP*.

Gu et al. (2021). Domain-specific LM pretraining for biomedical NLP. *ACM CHIL*.

Lee et al. (2020). BioBERT. *Bioinformatics*.

Yang et al. (2022). GatorTron. *npj Digital Medicine*.

Lehman et al. (2023). Do we still need clinical LMs? *CHIL*.

Labels & annotation:

Irvin et al. (2019). CheXpert. *AAAI*.

Ratner et al. (2017). Snorkel: rapid training data creation. *NeurIPS*.

Settles (2009). Active learning literature survey. *CS TR, U. Wisconsin*.

Zheng et al. (2023). Judging LLM-as-a-judge. *NeurIPS*.

Fairness & bias:

Sun et al. (2022). Negative patient descriptors. *J Gen Intern Med*.

Park et al. (2022). Stigmatizing language. *JAMA Netw. Open*.

Adekkanattu et al. (2024). Detecting stigma. *JAMIA*.

Zhang et al. (2020). Biases in clinical contextual word embeddings.

Factuality & evaluation:

Asgari et al. (2025). Hallucination in AI-generated notes. *npj Dig Med*.

Liu, Tsu et al. (2019). Clinically accurate CXR report generation. *MLHC*.

Tam et al. (2024). QUEST: Quality of health text evaluation framework. *NAACL*.

IAA benchmarks:

Uzuner et al. (2011). 2010 i2b2/VA challenge on concepts, assertions, relations. *JAMIA*.

Pang & Lee (2008). Opinion mining and sentiment analysis. *FTIR*.

Classical clinical NLP:

Chapman et al. (2001). NegEx. *AMIA*.

Harkema et al. (2009). ConText. *JBI*.

Datasets:

MIMIC-III/IV: physionet.org

MTSamples: kaggle.com

PubMed/PMC: pubmed.ncbi.nlm.nih.gov

Appendix: Classical Clinical NLP Pipeline

These approaches aren't the main focus of modern practice, but remain relevant: (a) many deployed systems still use them, (b) they're interpretable and auditable, and (c) they define the tasks LLMs are now benchmarked against.

Hospitals deployed NER / NegEx / RE systems for phenotyping, quality measurement, and clinical trials recruitment long before LLMs.

Notebook Figures 4–7.

Appendix: Clinical Named Entity Recognition (NER)

Task: Identify and classify mentions of clinical entities (conditions, medications, procedures, anatomy).

Standard pipeline:

$$\text{Raw text} \xrightarrow{\text{tokenize}} \text{Tokens} \xrightarrow{\text{NER}} \text{Entities} \xrightarrow{\text{link}} \text{UMLS CUIs}$$

Tool: scispaCy (pre-trained biomedical NER). Typical F1: 75–85% (vs. 90+ in general NER).

Why it's hard: Abbreviations, misspellings, context-dependence ("MS" $= 6$ meanings), nested entities.

Appendix: Negation Detection & Assertion Classification

Task: Determine whether an entity is present, absent, possible, conditional, or family history.

Classic approach — NegEx (Chapman et al., 2001): (1) Detect trigger terms ("no," "denies," "without"), (2) apply negation within a ∼6-word scope window, (3) mark entities in scope as negated. Achieves >90% specificity despite being purely rule-based.

Modern: Transformer-based assertion classifiers (ClinicalBERT), or LLM prompting.

Appendix: Relation Extraction

Task: Extract relationships between clinical entities.

- "Metformin treats diabetes" $\rightarrow$ (Metformin, TREATS, Diabetes)
- "CT showed pneumonia" $\rightarrow$ (CT, REVEALS, Pneumonia)

Approaches: Rule-based $\rightarrow$ supervised classifiers $\rightarrow$ LLM-based extraction.

Challenge: Each pipeline step adds error (NER errors $\times$ assertion errors $\times$ RE errors). LLMs can do end-to-end extraction, bypassing the pipeline.